BOOK PROPOSAL

Secure From the Start

Human-Centered Design, IT Auditing, and Threat Intelligence for Technology That Saves Money and Energy Before It Is Built

Anèl Henning

MS Cybersecurity Management candidate, Purdue University Global · BFA Graphic Design, MICA · BS Computer Science

Tampa, Florida · [email] · [phone] · Behance.net/anellehenning

Proposed length: about 70,000 words · 12 chapters · about 45 figures and tables

Manuscript delivery: 12 months after contract · Two sample chapters available

Status of underlying research: two peer-review manuscripts in preparation (2026)

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1. The Pitch

In one sentence

Most technology is secured, made accessible, and made efficient after it ships, when every fix is expensive; *Secure From the Start* shows designers, auditors, and small-business owners how to ask the right questions before building, using human-centered design, IT audit methods, and free threat intelligence.

A dentist who cannot read her own risk report. A grocery chain whose ordering system goes dark because a supplier's file-transfer server was hacked. A smart-home camera company fined because employees could watch customers' videos. A startup that promises a quantum-powered shopping cart. Each of these stories ends with money spent too late. This book teaches readers to spend a little thinking early instead.

The book is built on two research manuscripts by the author. The first applies a NIST-based IT audit toolkit and the CISA Known Exploited Vulnerabilities catalog to three small organizations: a dental group, a regional grocer, and a technical college. The second audits the author's own robotic grocery cart concept, Glidecart, before it is built, and tests its boldest claims with real experiments. Those experiments produced clear, surprising answers: quantum-safe encryption added only 262 microseconds per connection, ordinary route planning delivered the promised walking savings, and a simulated quantum algorithm lost to a classical one in every measure. Readers see the method, the numbers, and the reasoning, and then apply them to their own ideas.

2. Overview

Secure From the Start is a practical, visual guide to designing technology that people can trust and afford. It combines four disciplines that are usually taught separately:

Discipline	What the reader learns	Example from the book
Human-centered design	Start from the people who will use and maintain the product	Writing every audit finding twice: once for the auditor, once for the owner
IT auditing	Test every claim against evidence before trusting it	A 14-claim audit of the Glidecart concept: keep, revise, replace, or remove
Cyber threat intelligence	Use free, current data to decide what to fix first	Matching a store's equipment list to CISA's catalog of actively exploited products
Cost and energy thinking	Choose low-cost, low-energy options while they are still free to choose	Quantum-safe encryption costs about 8 watt-hours a year for a 200-cart fleet when built in from day one

The book is organized in four parts: **Start With People**, **Know Your Threats**, **Design Smart Devices That Earn Trust**, and **Prove It**. Each chapter ends with a one-page checklist and a short "Try It" exercise that uses only free tools.

What the book is not: it is not a hype book about quantum computing or robots, and it is not a compliance manual. It treats bold technology claims the way an auditor would, with respect and with a request for evidence.

3. Why This Book, Why Now

- **Breaches keep getting more expensive.** The average data breach cost USD 4.88 million in 2024 (IBM, 2024). Small organizations that cannot absorb that cost are a growing share of victims.
- **Regulators now punish design failures.** In 2023 the FTC settled with Ring over employee access to customer videos and with Amazon over retained Alexa voice recordings of children. These were design choices, not hacks.
- **Quantum is arriving in two different ways.** NIST published post-quantum encryption standards in 2024 and proposes retiring today's public-key algorithms between 2030 and 2035, while quantum computing is also widely oversold. Readers need to know which half to act on now.
- **Retail and healthcare are automating fast.** Smart carts, deli kiosks, voice assistants, and patient portals are being deployed by organizations with no security staff.
- **Secure by design is now official policy.** CISA and its international partners ask manufacturers to build security in by default. Designers and small vendors need a plain-language way to do it.

4. Audience

Reader	What they need	How the book serves them
Small-business owners and IT generalists (dental, medical, retail, schools)	Know what to fix first with no security team	Checklists, plain-language risk registers, free threat-intelligence steps
Product designers, UX researchers,	Build trust, accessibility, and	Design-thinking stages mapped to

and founders	efficiency into new devices	security and energy decisions; Glidecart case
IT audit, GRC, and cybersecurity students	See audit methods applied to real cases	Worked cases, labs, and discussion questions; suitable as a course text
Instructors in cybersecurity, IS, and design programs	Ready-made teaching material	A 90-minute companion lesson, lab scripts, and instructor notes

Primary market: practitioners and students in the United States, with relevance to the EU, Canada, and Japan through the book's accessibility and privacy mapping (GDPR, PIPEDA, APPI, JIS X 8341, CSA B651). **Course adoption:** graduate and upper-level undergraduate courses in IT audit, security risk management, usable security, and design for emerging technology.

5. Comparable and Competing Titles

No current title combines human-centered design, IT auditing, threat intelligence, and cost and energy decisions in one practical book for small organizations and product teams. The closest titles each cover one or two of these areas.

Title	Strength	How *Secure From the Start* differs
Norman, *The Design of Everyday Things* (Basic Books, rev. ed. 2013)	Classic on human-centered design	Applies the same principles to security, audit, and energy decisions
Brown, *Change by Design* (Harper Business, 2009)	Popularized design thinking in business	Adds risk, evidence, and threat data to the design-thinking cycle
Shostack, *Threat Modeling: Designing for Security* (Wiley, 2014)	Definitive threat-modeling guide for engineers	Written for non-specialists; adds audit, cost, and user-research methods
Cranor & Garfinkel (Eds.), *Security and Usability* (O'Reilly, 2005)	Founding usable-security collection	Current threats, current regulation, and step-by-step practice
Adkins et al., *Building Secure and Reliable Systems* (O'Reilly, 2020)	Large-scale engineering practice	Aimed at small organizations and early-stage concepts
Schneier, *Click Here to Kill Everybody* (W. W. Norton, 2018)	Policy case for securing connected devices	Gives designers and owners concrete methods, not only policy
Bernhardt, *Quantum Computing for Everyone* (MIT Press, 2019)	Accessible introduction to quantum computing	Focuses on what to do now: post-quantum migration and testing claims

6. About the Author

Anèl Henning is a cybersecurity management graduate student at Purdue University Global and a designer with about fifteen years of experience in multichannel communication for healthcare, corporate, and nonprofit clients. She holds a BFA in Graphic Design from the Maryland Institute College of Art and a BS in Computer Science from Rasmussen College. At the Moffitt Cancer Center she supervised creative services and designed the “Breathe Deep” awareness campaign for the Thoracic Department. Since 2023 she has worked in business development and content support at Logical Coders in Tampa, where she built a browser-based governance, risk, and compliance toolkit for small healthcare practices. Her years in grocery retail inform the book's retail cases.

Why she is the right author: the book sits where her three backgrounds meet. Designers rarely write about audit evidence, and auditors rarely write about visual communication; Henning has trained in

both and has built the tools the book describes. She is preparing for the ISC2 Certified in Cybersecurity and CompTIA Security+ certifications.

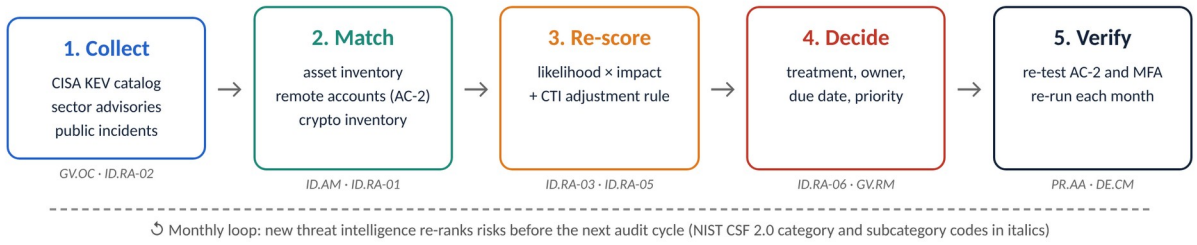
Research: two manuscripts in preparation for peer review (2026): a teaching case on threat intelligence for small-organization audits, and a design-science evaluation of the Glidecart concept with computational experiments. Both are the foundation for this book.

Platform: design portfolio on Behance; public GRC toolkit and case studies on GitHub Pages; LinkedIn; graduate cybersecurity community. The author plans talks at regional events such as local ISACA and WiCyS chapters and BSides Tampa.

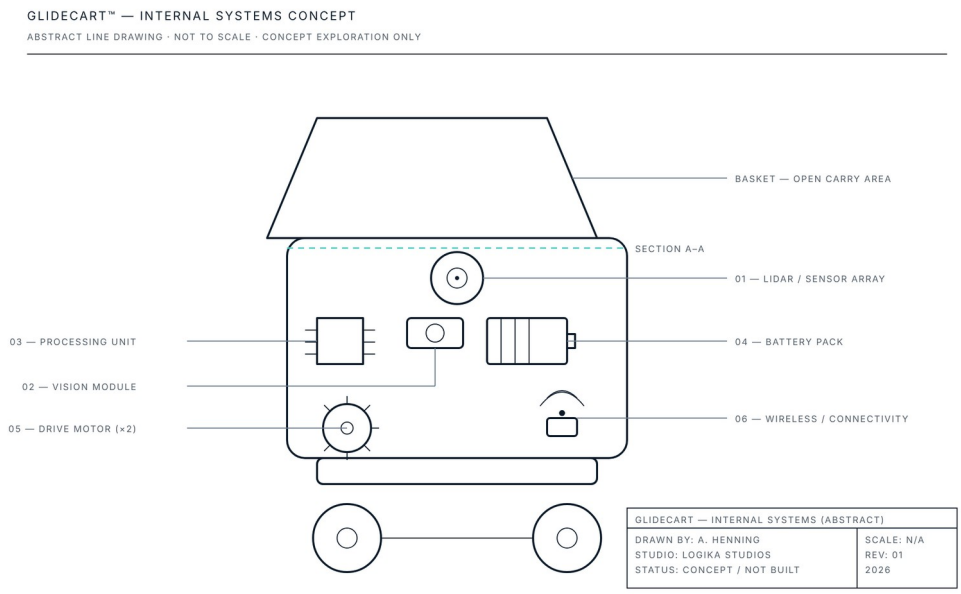
7. Book Features and Design

- **Visual by design.** About 45 original figures and tables in a consistent white-page, colored-outline style designed by the author, including risk heat maps, process loops, and annotated device drawings such as the Glidecart internal-systems diagram.
- **Worked cases.** Five recurring cases: Sunshine Dental Group, Bayside Fresh Market, Pelican Bay Technical College, the Glidecart cart, and a deli ordering kiosk. All organizations are clearly labeled as fictional teaching cases anchored to real public incidents.
- **Real numbers.** Threat data from the CISA catalog and results from reproducible experiments, with code available in a companion repository.
- **Plain language.** Each technical idea is explained once for a business owner and once for a practitioner.
- **End-of-chapter tools.** A one-page checklist, a “Try It” exercise with free tools, and discussion questions for classrooms.
- **Honest reporting.** The book shows experiments that disproved the author's own early claims, modeling how to report results that do not go your way.

Sample visuals from the research manuscripts:



The five-step threat-intelligence loop (Chapter 4).



Glidecart internal-systems concept drawing, used for the cyber-physical threat model (Chapter 8).

8. Chapter Outline

Part I · Start With People

Chapter 1 The Expensive Afterthought

Why security, accessibility, and energy efficiency cost the most when added last. The breach-cost data, the cost-of-change argument, and the Ring and Alexa settlements as design failures rather than hacks. Introduces the five cases.

Approx. words: 4,500 · Paper 2

Chapter 2 Design Thinking Meets Risk

The five design-thinking stages (empathize, define, ideate, prototype, test) mapped to audit and security work. How to write findings an owner can act on, and why color, layout, and plain language are security controls.

Approx. words: 6,000 · Paper 1

Chapter 3 Audit the Idea Before You Build It

The claims-to-evidence method: turn every promise into a hypothesis, test it, and decide to keep, revise, replace, or remove it. Worked example: the 14-claim audit of the Glidecart and retail-robotics concepts, including an arithmetic error and a misidentified chip.

Approx. words: 6,000 · Paper 2 (sample chapter)

Part II · Know Your Threats

Chapter 4 Threat Intelligence Without an Analyst

Using the free CISA Known Exploited Vulnerabilities catalog to rank risks. The five-step loop: collect, match, re-score, decide, verify. Why a missing match is not proof of safety, and why consumer devices need other sources.

Approx. words: 6,000 · Paper 1 (sample chapter)

Chapter 5 Three Small Organizations

The dental group, the grocer, and the technical college: inventories, KEV matches, MFA gaps, and before-and-after risk scores, each tied to a real incident (Change Healthcare, Ahold Delhaize and UNFI, PowerSchool).

Approx. words: 7,000 · Paper 1

Chapter 6 Accounts, MFA, and the People Behind Breaches

Account management (NIST AC-2) as the most common root cause in the cases. A Python and SQL account review anyone can run, and how to make MFA usable rather than resented.

Approx. words: 5,000 · Paper 1 and capstone

Part III · Design Smart Devices That Earn Trust

Chapter 7 Voice, Cameras, and the Smart-Home Lesson

What Ring doorbells and Alexa speakers taught us. Inaudible and laser-injected voice commands. Ten design requirements for voice and touch, each with the money or energy it saves: on-device processing, voice that cannot pay, privacy lights, and more.

Approx. words: 6,000 · Paper 2

Chapter 8 Robots in the Aisle

Threat modeling Glidecart's six internal systems: LiDAR, vision, processor, battery, motors, and wireless. Cyber-physical safety, food safety at the deli counter, touchless ordering, and why a biometric “impulse detector” was removed from the design.

Approx. words: 6,500 · Papers 1 and 2

Chapter 9 Quantum: Hype, Hope, and What to Do Now

The experiments in plain language: a simulated quantum routing algorithm found the best route in 13 of 30 store layouts while a simple classical search was always right; Grover search and quantum recommendation re-examined. Then the urgent part: migrating to post-quantum encryption before 2030.

Approx. words: 6,500 · Paper 2

Chapter 10 Saving Energy From the Start

Energy as a design requirement. Quantum-safe encryption measured at 262 microseconds per connection, sleep modes, on-device processing, event-triggered cameras, and how to measure energy per shopping trip.

Approx. words: 5,000 · Paper 2 and new research

Part IV · Prove It

Chapter 11 Hypotheses, Pilots, and Honest Numbers

How to design small studies: the voice-versus-touch usability study (34 participants) and the deli wait-time field test (98 orders per condition). Power analysis, ethics review, pre-registration, and reporting results that disprove your idea.

Approx. words: 5,500 · Paper 2

Chapter 12 From Concept to Store

An evidence-gated roadmap where each stage proves value before the next spends more: under USD 200, then 2,000, then 5,000, then a hardware partnership. Ends with the complete Secure From the Start checklist.

Approx. words: 4,500 · Paper 2

Appendices Toolkit, Scripts, and Glossary

Risk-register and gap-analysis templates, the threat-intelligence and experiment scripts, a 90-minute companion lesson for instructors, and a plain-language glossary.

Approx. words: Variable

9. Sample Chapters

Two sample chapters can be delivered within eight weeks of a publisher's request:

- **Chapter 3, Audit the Idea Before You Build It.** Shows the book's signature method and its honesty: the author audits her own concept and reverses several of its claims.
- **Chapter 4, Threat Intelligence Without an Analyst.** Shows the practical, step-by-step style with a free data source and a working script.

Both chapters are adapted from the research manuscripts, rewritten for a general professional audience, and illustrated in the book's visual style.

10. Research Foundation

Manuscript	What it contributes to the book	Key results
Manuscript 1: Human-centered cyber threat intelligence for small-organization audits (teaching case, in preparation)	Chapters 2, 4, 5, 6, and the instructor materials	Across three cases, 10 of 18 assets were internet-facing with recent ransomware-linked KEV entries; 10 of 15 remote accounts lacked MFA; 9 of 11 risks rose after threat intelligence
Manuscript 2: Secure, human-centered, and energy-aware from the start (design-science study, in preparation)	Chapters 1, 3, 7, 8, 9, 10, 11, 12	Hybrid post-quantum key exchange added 262 µs per session; route optimization saved 18.4% walking distance; simulated QAOA found the optimal route in 13 of 30 layouts; 14 concept claims audited

All organizations in the cases are fictional and labeled as such. Threat data come from the public CISA catalog (version 2026.10.08). Experiments are reproducible with the scripts in the companion repository. The book will update the threat data and, where completed, add results of the planned user studies.

11. Marketing and Promotion

- **Course adoption:** outreach to instructors of IT audit, GRC, usable security, and design programs, supported by the companion lesson, slides, and lab scripts.
- **Companion repository:** free templates and scripts on GitHub that point readers back to the book.

- **Talks and workshops:** proposals to regional ISACA and WiCyS chapters, BSides Tampa, and small-business and healthcare associations in Florida.
- **Articles:** short LinkedIn and industry-blog articles drawn from each chapter, for example “What Ring and Alexa Teach Product Designers” and “The 8-Watt-Hour Case for Quantum-Safe Encryption.”
- **Visual assets:** the author will create social graphics and a one-page visual summary of each chapter in the book's design style.
- **Peer-reviewed publications:** the two research manuscripts, once published, will cite and promote the book.

12. Specifications and Schedule

Item	Detail
Length	About 70,000 words plus appendices
Illustrations	About 45 original figures and tables, supplied by the author as print-ready files
Format	Print and e-book; companion repository for templates and scripts
Permissions	No third-party images required; all figures are original
Sample chapters	Within 8 weeks of request
Full manuscript	12 months after contract
Milestones	Part I at month 3 · Part II at month 6 · Part III at month 9 · Part IV and appendices at month 12

13. Target Publishers

Each publisher has its own proposal form; this document supplies the content for all of them. Submit to one or a few at a time and follow each publisher's simultaneous-submission policy.

Publisher	Why it fits
Rosenfeld Media	Specializes in UX and design books for practitioners; strongest fit for the human-centered angle
Apress (Springer Nature)	Practical technology books; open to first-time authors in security and design
CRC Press / Auerbach (Taylor & Francis)	Large list of security, audit, and GRC titles; strong course-adoption channel
Wiley	Publishes major security titles, including threat modeling
SpringerBriefs (Springer)	Short academic books (about 50 to 125 pages); a faster first book based on the two manuscripts

14. Query Letter

Dear [Acquisitions Editor's name],

I am writing to propose *Secure From the Start: Human-Centered Design, IT Auditing, and Threat Intelligence for Technology That Saves Money and Energy Before It Is Built*, a practical, illustrated book of about 70,000 words for designers, small-business owners, and cybersecurity students.

Most technology is secured, made accessible, and made efficient after it ships, when every fix is expensive. My book shows readers how to ask the right questions first, using design thinking, audit methods, and free threat intelligence. It is built on two research manuscripts of mine, including experiments on my own robotic grocery-cart concept: quantum-safe encryption cost only 262 microseconds per connection, while a simulated quantum routing algorithm lost to a simple classical method. Readers see the method, the numbers, and the reasoning, and then apply them to their own ideas.

I bring together three backgrounds that rarely meet: a BFA in graphic design from MICA with about fifteen years of design work, including healthcare campaigns at Moffitt Cancer Center; a BS in computer science; and graduate study in cybersecurity management at Purdue University Global. I will supply all figures in a consistent, print-ready style.

The full proposal, chapter outline, and two sample chapters are available on request. Thank you for your consideration.

Sincerely,

Anèl Henning

[email] · [phone] · Behance.net/anellehenning